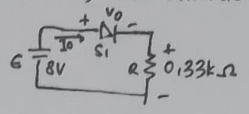
ELEKTRONIKA ANALOG

Nama : Ailsa Rahma
NIM : 24133100 2
kelas : 1A TTE

BAB 2

* 2.2 Load line Analysis

1). a). using the characteristic of Fig 2.131b, determine I_D, V_D and V_R for the circuit of Fig 2.131a



$V_{DQ} = 0,85\ V$

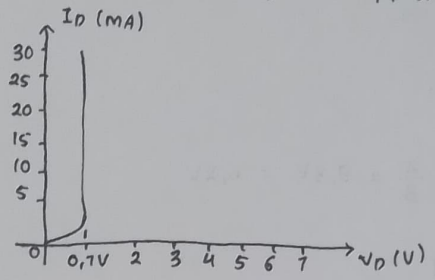
$I_{DQ} = 15\ mA$

$V_R = E - V_{DQ} = 12V - 0,85V = 11,15\ V$

$I_D = \frac{E}{R} = \frac{12V}{150\Omega} = 16mA$

$V_D = 12\ V$

b). Repeat part a using the approximate model for the diode and compare results



$V_{DQ} = 0,7\ V$

$I_{DQ} = 15\ mA$

$V_R = E - V_{DQ} = 12V - 0,7V = 11,3\ V$

c). Repeat part a using the ideal model for the diode and compare results

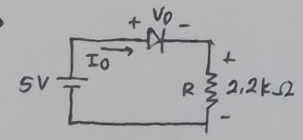
$V_{DQ} = 0\ V$

$I_{DQ} = 16\ mA$

$V_R = E - V_{DQ} = 12V - 0V = 12\ V$

3). Determine the value of R for the circuit of Fig 2.132 that will result in a diode current of 10 mA if $E = 7V$. use the characteristics of Fig 2.132b for the diode

↳



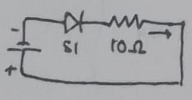
$I_D = 11,3\ mA = \frac{E}{R} = \frac{7V}{R}$

With $R = \frac{7V}{11,3\ mA} = 619,47k\Omega \cong 0,62k\Omega$

* 2.4 Series Diode Configurations w/DC Inputs

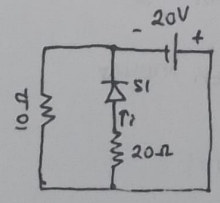
5). Determine the current I for each of the configurations of Fig 2.134 using the approximate equivalent model for the diode

↳ a).



$I = 0mA$, diode reserve bias

b).



$V_{20\Omega} = 20 - 0,7V = 19,3\ V$

$I(20\Omega) = \frac{19,3V}{20\Omega} = 0,965\ A$

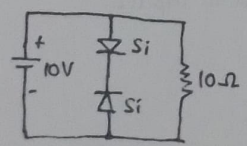
$V(10\Omega) = 20V - 0,7V = 19,3\ V$

$I(10\Omega) = \frac{19,3V}{10\Omega} = 1,93A$

$I = I(10\Omega) + I(20\Omega)$

$= 2,895\ A$

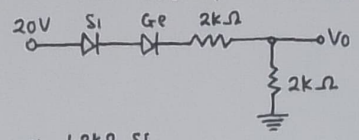
c).



$I = \frac{10V}{10\Omega} = 1A$

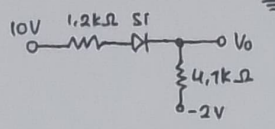
7). Determine the level of V_o for each network

a).



$$V_o = \frac{10k\Omega\,(12V - 0,7V - 0,3V)}{2k\Omega + 10k\Omega} = 9,17V$$

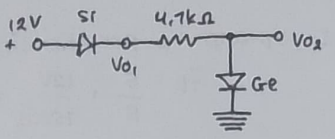
b).



$V_o = 10V$

9). Determine V_{o_1} and V_{o_2} for the network

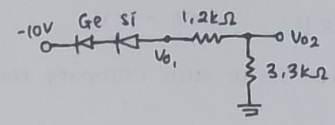
a).



$V_{o_1} = 12V - 0,7V = 11,3V$

$V_{o_2} = 1,2V$

b).

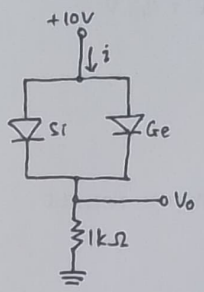


$V_{o_1} = 0V$

$V_{o_2} = 0V$

11). Determine V_o and I for the networks

a).

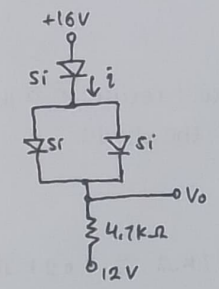


$$V_o = \frac{2k\Omega\,(10V - 0,7V)}{1k\Omega + 2k\Omega} = \frac{2}{3} = 9,3V = 6,2V$$

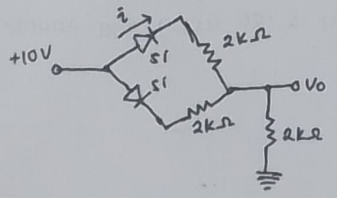
$$I_{2k\Omega} = \frac{6,2V}{2k\Omega} = 3,1mA$$

$$I_D = \frac{I_{2k\Omega}}{2} = \frac{3,1}{2} = 1,55\,mA$$

b).



13). Determine V_o and I_D for the network



$V_{o_1} = 0,7V$, $V_{o_2} = 0,3V$

$$I_{1k\Omega} = \frac{20V - 0,7V}{1k\Omega} = \frac{19,3V}{1k\Omega} = 19,3mA$$

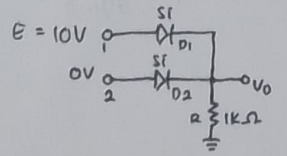
$$I_{0,47k\Omega} = \frac{0,7V - 0,3V}{0,47k\Omega} = 0,851\,mA$$

$I(Si) = I_{1k\Omega} - I_{0,47k\Omega}$

$= 19,3 - 0,851$

$= 18,45\,mA$

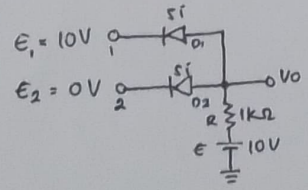
15). Determine V_o for the network of Fig 2.38 with 10 V on both inputs



$V_o = 10V - 0,7V$

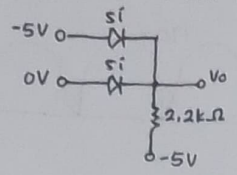
$= 9,3V$ (kedua dioda menyala)

17). Determine V_o for the network of Fig 2.41 with 10V on both inputs



$V_o = 10V$ (kedua dioda mati)

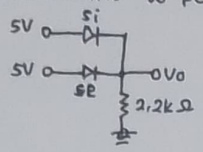
19). Determine V_o for the negative logic AND gate of Fig 2.144



$V_o = 0V - 0,7V$

$= -0,7V$

21). Determine V_o for the configurations of Fig 2.146



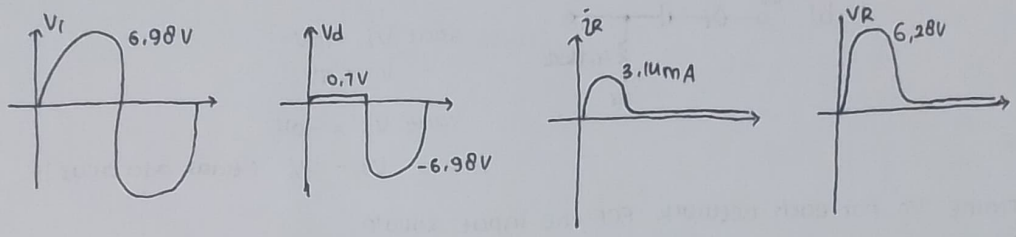
$V_o = 5V - 0,3V$

$= 4,7V$

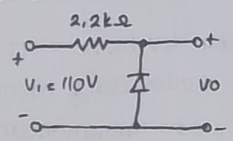
23). Repeat problem 22 with a silicon diode ($V_T = 0,7V$)

↳ Dik : $V_{dc} = 0,318\ (V_m - V_T)$ $\quad V_m = 6,98\,V$

$= 0,318\ (V_m - 0,7V)$



25). For the network of Fig 2.149 sketch V_o and determine V_{dc}

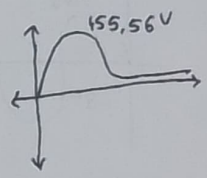


$V_m = \sqrt{2}\ (110) = 155,56\,V$

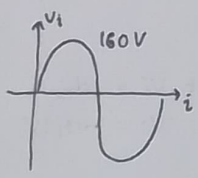
$V_{dc} = 0,318 \cdot V_m$

$= 0,318\ (155,56\,V)$

$= 49,46\,V$



27). a).



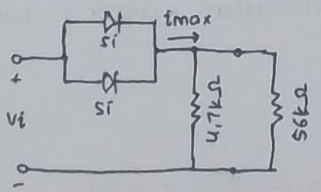
given $P_{max} = 14\,mW$, determine the max current rating of each diode (using the approximate equivalent model)

↳ $I_D = \frac{14mW}{0,7V} = 20mA$

b). Determine I_{max} for $V_{Imax} = 160\,V$

↳ $I_{max} = 2 \times 20mA = 40mA$

c). Determine the current through each diode at V_{Imax} using the results of part b



$V_R = 160V - 0,7V = 159,3\,V$

$4,7k\Omega \,||\, 68k\Omega = 4,4k\Omega$

$I_{max} = \frac{159,3V}{4,4k\Omega} = 36,2\,mA$

$I_d = \frac{I_{max}}{2}$

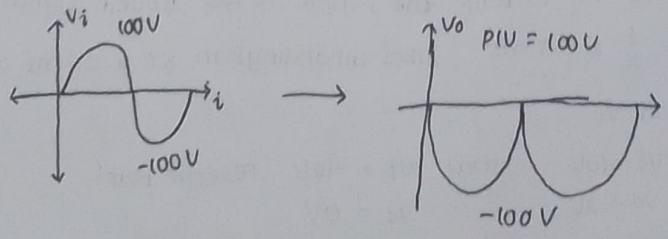
$= \frac{36,2}{2} = 18,1\,mA$

d). If only one diode were present, determine the diode current and compare it to the max rating

↳ kerusakan total = $36,2mA > 20mA$

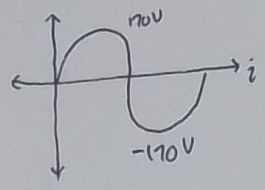
✱ 2.8 Full wave Rectification

29). Determine V_o and the required PIV rating of each diode for the configuration

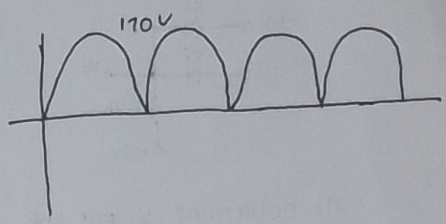


$I_{max} = \frac{100V}{2.2k\Omega} = 45.45\,mA$

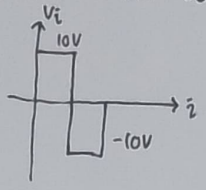
31). Sketch V_o for the network and determine the dc voltage available



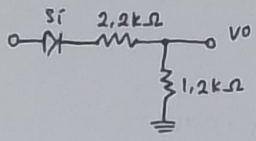
$$V_{dc} = \frac{2V_m}{\pi} = \frac{2(170V)}{\pi} \approx 108,1\,V$$



33). Determine V_o for each network for the input shown



a).



saat $V_i = 10V$

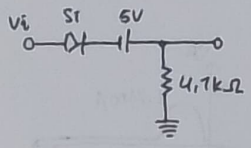
$V_o = V_i - 0,7V$

$= 10V - 0,7V = 9,3V$

saat $V_i = -10V$ (arus tidak mengalir)

$V_o = 0V$

b).



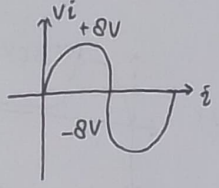
saat $V_i = 10V$

$V_o = 5V$

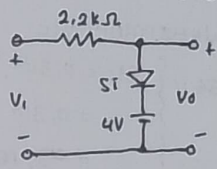
saat $V_i = -10V$

$V_o = 0V$ (tidak ada arus)

35). Determine V_o for each network for the input shown



a).



saat $V_i = 8V$

V_i lebih besar dari 4V + 0,7V = 4,7 V

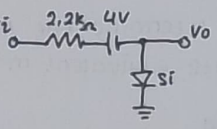
Jika $V_i < 4,7V$, dioda tidak menghantarkan

Jika $V_i > 4,7V$, dioda menghantar, $V_o = 4V$

saat $V_i = -8V$

$V_o = 0V$

b).



saat $V_i = 8V$

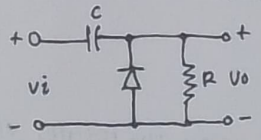
$V_o = 4V$

saat $V_i = -8V$

$V_o = -0,7V$

37). Sketch V_o for each network for the input shown

a).



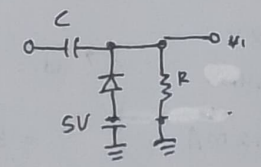
saat $V_i = +20V$

V_o = karena dioda ideal maka $V_o = 0V$ (Gnd)

saat $V_i = -20V$

V_o = tegangan output digeser ke bawah sebesar -40V

b).



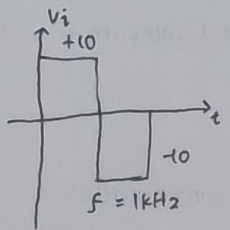
saat $V_i = +20V$

$V_o = 5V$

saat $V_i = -20V$

$V_o = V_i - 5V = (-20) - 5 = -25V$

39).



a). calculate $5T$

$\tau = RC$

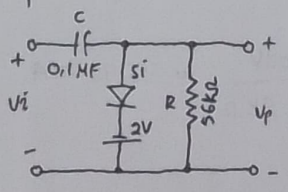
$= (56 \times 10^3) \times (0,1 \times 10^{-6})$

$= 5,6 \times 10^{-3}\,s = 5,6\,ms$

$5T = 5 \times 5,6ms = 28\,ms$

b). compare $5T$ to half the period of the applied signal

$\frac{T}{2} = \frac{1}{2} = 0,5\,ms$, jika dibandingkan $5T = 28ms > 0,5ms$



c). sketch V_o

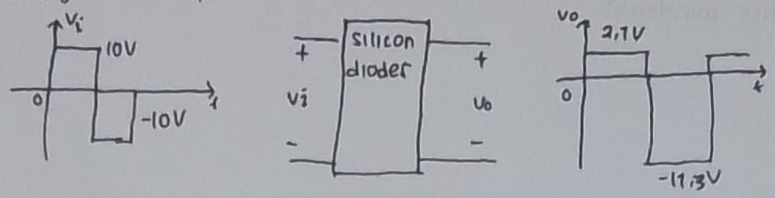
saat $V_i = 10V$

$V_o = 2V$

saat $V_i = -10V$ (reserve bias)

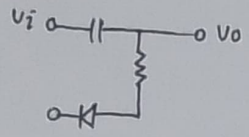
$V_o = 0V$

41). Design a clamper to perform the function indicated

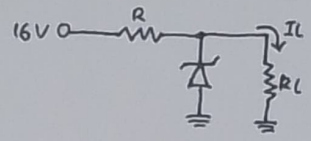


Pergeseran = 10V - (-17,3V) = -7,3V

negative clamper



43). a). Design the network to maintain V_L at 12V for a load variation (I_L) from 0 to 200 mA. That is, determine R_S and V_Z



$V_S - V_L = 16V - 12V = 4V$

$I_S = 200mA + 20mA = 220mA$

$R_S = \frac{4V}{220mA} = 18,2\ \Omega$

$I_{Zmax} = \frac{4V}{18\Omega} = 222mA$

Jadi $R_S = 18\Omega$ dan $V_Z = 12V$

b). Determine P_{Zmax} for zener diode of part a

$P_{Zmax} = V_Z \times I_{Zmax}$

$= 12V \times 222mA = 2,66\ W$

45). Design a voltage regulator that will maintain an output voltage of 20V across a 1-kΩ load with an input that will vary between 30 and 50 V. That is, determine the proper value of R_S and the max current I_{ZM}

↳ • $I_L = \frac{V_L}{R_L} = \frac{20V}{1k\Omega} = 20mA$

• $R_S = \frac{V_S - V_Z}{I_S}$ → I_S = total arus $= I_L + I_Z$

• Tegangan input minimum ($V_S = 30V$)

$I_S = I_L + I_Z$

$= 20mA + 5mA = 25mA$

$R_S = \frac{30V - 20V}{25mA} = \frac{10V}{25mA} = 400\Omega$

→ tegangan input max ($V_S = 50V$)

$I_S = \frac{50V - 20V}{400\Omega} = \frac{30V}{400\Omega} = 75mA$

$I_{Zmax} = I_S - I_L$

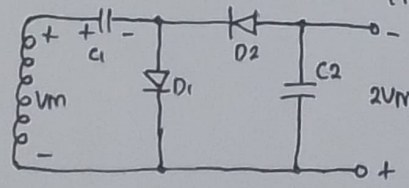
$= 75mA - 20mA = 55mA$

• daya max dioda zener

$P_{Zmax} = V_Z \times I_{Zmax}$

$= 20V \times 55mA = 1,1\ W$

47). Determine the voltage available from the voltage doubler if the secondary voltage of the transformer is 120 V (rms)



$V_p = V_{rms} \times \sqrt{2}$

$= 120V \times 1.414 = 169,7V$

$V_{out} = 2 \times V_p$

$2 \times 169,7V = 339,4V$